

divide by larger array as p12
can also be solved/proved with
tree or master theorem

LECTURE-08

Monday
14/9/20

□ Proving Best Case of Binary Search by Substitution:

$$T(n) = T(n/3) + \mathcal{O}(1) \longrightarrow T(n) = \mathcal{O}(\log_3 n)$$

$$T(n) \leq C \cdot \log(n/3) + C$$

★ RHS provides upper bound to LHS ($T(n)$).

$$T(n) \leq C \cdot \log_3(n) + C \log_3\left(\frac{1}{3}\right) + C$$

$$T(n) \leq C \cdot \log_3(n) + \cancel{C(-1)} + \cancel{C}$$

$$T(n) \leq C \cdot \log_3(n)$$

★ $T(n)$ needs to be '1' and C always greater equal to 1.

So for that ; 'n' needs to be '3' as $\log_3$ cancel with $n=3$

$$C \geq 1, \quad n = 3$$

→ * Put $(n-1)$ in place of n in n^2 as guess ~~is~~ $\rightarrow O(n^2)$.

Worst Case of Binary Search $\rightarrow O(n^2)$

$$T(n) = T(n-1) + O(n)$$

$$T(n) = C \cdot n^2 + cn \rightarrow O(n^2)$$

* plug $(n-1)$ at 'n' in ① which is for $T(n-1)$

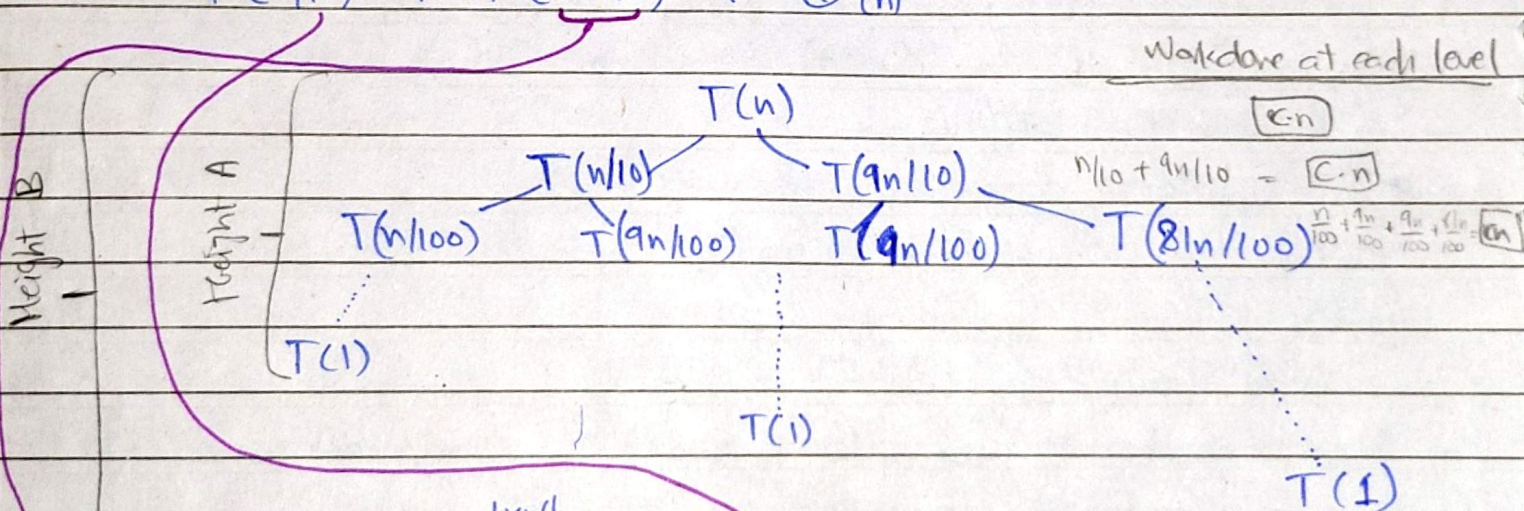
$$T(n) = C \cdot (n-1)^2 + cn$$

$$T(n) = cn^2 - 2cn + c + cn$$

$$T(n) = cn^2 - cn + c$$

Average Case of QUICK SORT: ^{with tree} 90% in smaller array
10% in larger array.

$$T(n) = T(n/10) + T(9n/10) + O(n)$$



Height A = smallest $\uparrow = \log_b(n) = \log_{10}(n)$

Height B = largest Branch = $\log_b(n) = \log_{10/9}(n)$

$T(n) =$ WD at terminal Nodes * Average no. of terminal

we'll just
add smallest & largest branch.
Date _____

$T(n) = (\text{WD at terminal nodes}) * \text{Avg. no. of terminal nodes in each level}$

$$T(n) = \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$
$$T(n) = cn \left[\log_{10}(n) + \log_{10/9}(cn) \right]$$

$$\boxed{T(n) = \Theta(n \log n)}$$

NOTE: we didn't considered terminal nodes in $T(n)$ as we can't compute exact no. of 'k' since diff heights & the effect of it will still take us in " $n \log n$ ".